consider only the case of Wiener increments and delta-correlated (memory-less) processes.

Langevin's idea of including external random impacts in the equations of motion for a free particle can be extended to a system of particles. The equations of motion are, in vectorial form:

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{6.32}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma\boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma \mathrm{k_B T}}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}. \tag{6.33}$$

The physical interpretation is that we have a system of particles immersed in a fluid bath consisting of many particles which are much lighter (or give rise to weaker interactions) than those described by the potential U.

The coefficient γ, referred to alternatively as the *friction coefficient* or the *collision rate*, is a frequency and therefore has units of $1/\text{time}$.

It is possible to further extend this model in a number of ways. In particular, a different friction coefficient γ can be used for each particle and these coefficients can even depend on spatial location, thus we might work with the following system:

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{6.34}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{\Gamma}(\boldsymbol{q})\boldsymbol{\Gamma}(\boldsymbol{q})^T\boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}(\boldsymbol{q})\mathrm{d}\hat{\mathbf{W}}, \tag{6.35}$$

where we use the matrix $\Gamma(\boldsymbol{q}) \in \mathbb{R}^{\mathrm{N_c}\times S}$ and a vector of S independent Wiener processes $\hat{\mathbf{W}}$.

In the case of Langevin dynamics with different, variable friction coefficients for each particle, we have $S = \mathrm{N_c}$ and

$$\Gamma = \mathrm{diag}(\gamma_1(\boldsymbol{q}), \gamma_1(\boldsymbol{q}), \gamma_1(\boldsymbol{q}), \gamma_2(\boldsymbol{q}), \gamma_2(\boldsymbol{q}), \gamma_2(\boldsymbol{q}), \ldots, \gamma_{\mathrm{N}}(\boldsymbol{q}), \gamma_{\mathrm{N}}(\boldsymbol{q}), \gamma_{\mathrm{N}}(\boldsymbol{q})).$$

Another application is *dissipative particle dynamics* [181], where typically $S = \mathrm{N_c}(\mathrm{N_c} - 1)/2$ and a random increment is applied for each pair interaction.

The Langevin dynamics equations have the Markov property in the sense that the state of the system at any future time has a dependency on the current state of the system, but not on the prior history.

Langevin dynamics can be regarded as a physical model where the parameter-functions encoded in $\boldsymbol{\Gamma}(\boldsymbol{q})$ describe the instantaneous physical friction forces between the modelled particles and a simplified, anisotropic bath of particles. However, it should be noted that in many types of modelling, the physical nature of the interaction with the solvent bath is neglected and γ or $\boldsymbol{\Gamma}(\boldsymbol{q})$ are simply viewed as artificial parameters which allow recovery of averages with respect to the canonical distribution.